Retrieval-Augmented Generation (RAG) → Perplexity AI

traditional LLMs

trained on training data

Query → LLM → output generate

improve the LLM output

1. up to date
2. accuracy ↑
3. fact based ↑

1. Hallucination – making things up
2. Not upto date

Information Retrieval

APNA COLLEGE

RAG

Architecture

RAG architecture

Data Ingestion → Embeddings → store in vector DB

Query → Query Processing → vector DB

Retrieval → Augmentation → LLM → generation

vector Database

Retrieve

Query → LLM → output generate

SQL - traditional DB based

name = "Harry Potter"

vector DB

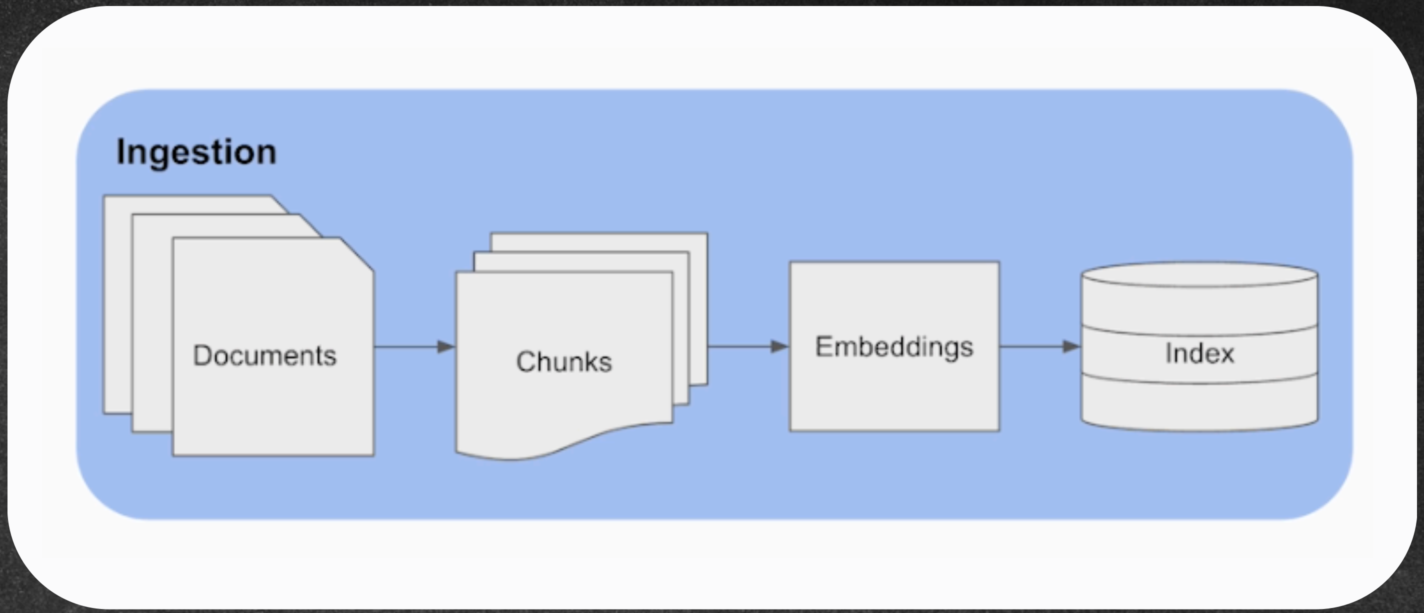
→ wizards, children fairystory

RAG

Data Ingestion Pipeline

context size

LLM



text → nums

vector embeddings

↑

① Data Ingestion → ② Embedding → ③ Vector DB

→ Knowledge Base

pdf
html
db
excel
⋮

⇓

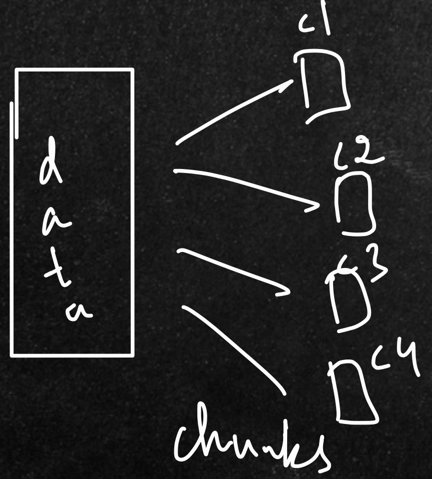
process/clean

⇓

chunks

- OpenAI embedding models
- sentence transformer

Pinecone
FAISS
chromadb
⋮



RAG

Retrieval Pipeline

semantic / similarity search

5 Retrieval

vector DB

4 Query Processing

Query

Query Embeddings

top K similar results

context

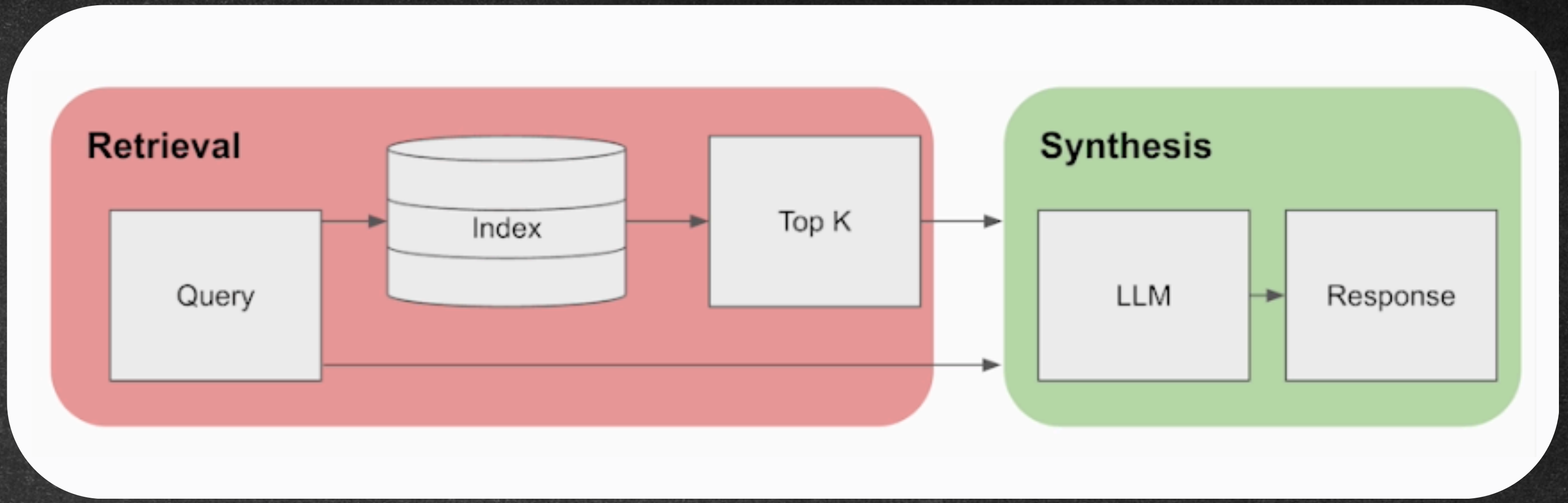
6 Augmentation

+ prompt

LLM

o/p generate

7 Generation



- Nearest neighbour search

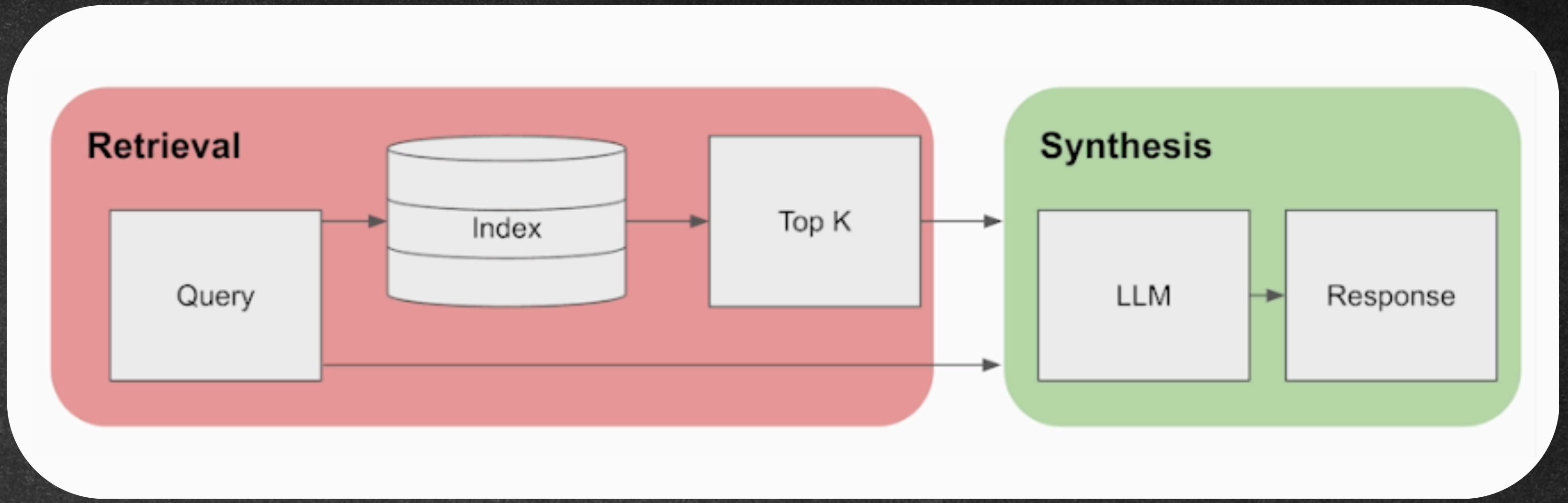
- Cosine Similarity (vectors) ↑

cosine dist ↓

cosine similarity = 1 − cosine dist

RAG

Retrieval Pipeline



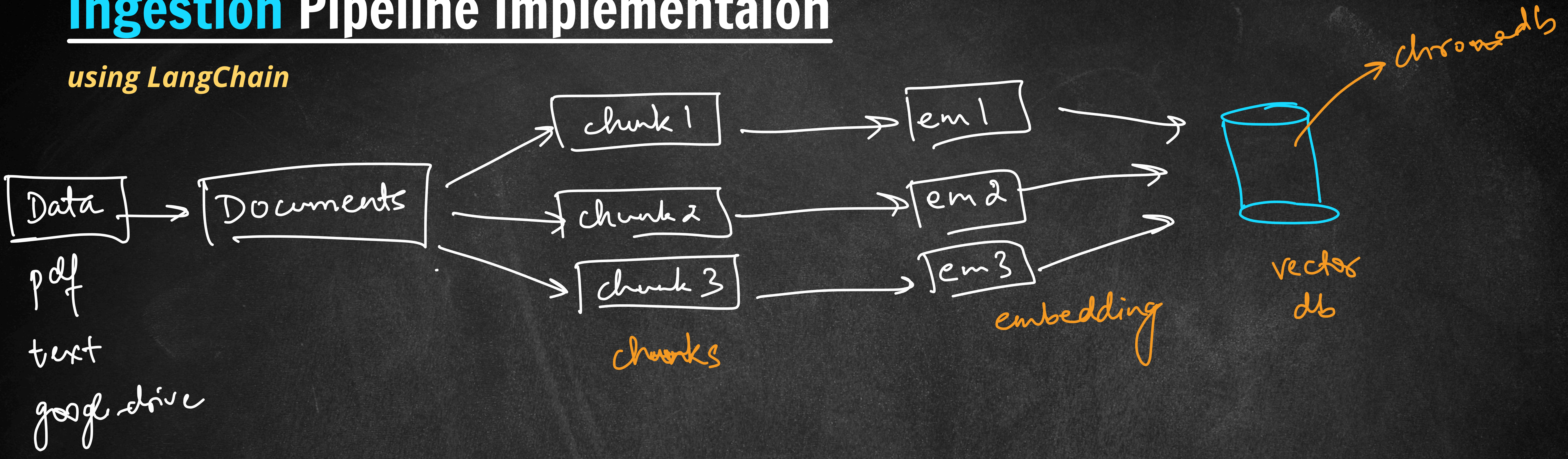
doc1: programming
doc2: fruits
doc3: politics

Query Processing → Retrieval → Augmentation → Generation

Python

Ingestion Pipeline Implementaion

using LangChain



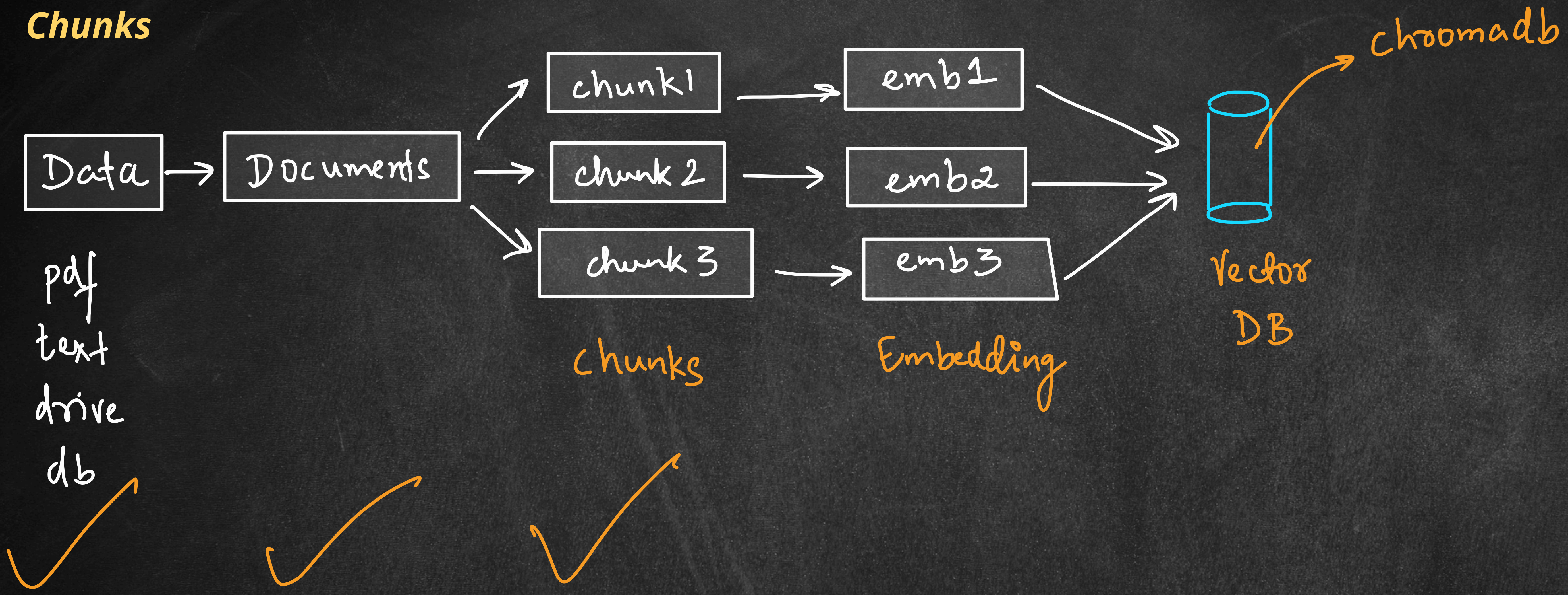
(1) Document → object

{
(1) text/ page-content
(2) metadata
}

(2) Document loaders

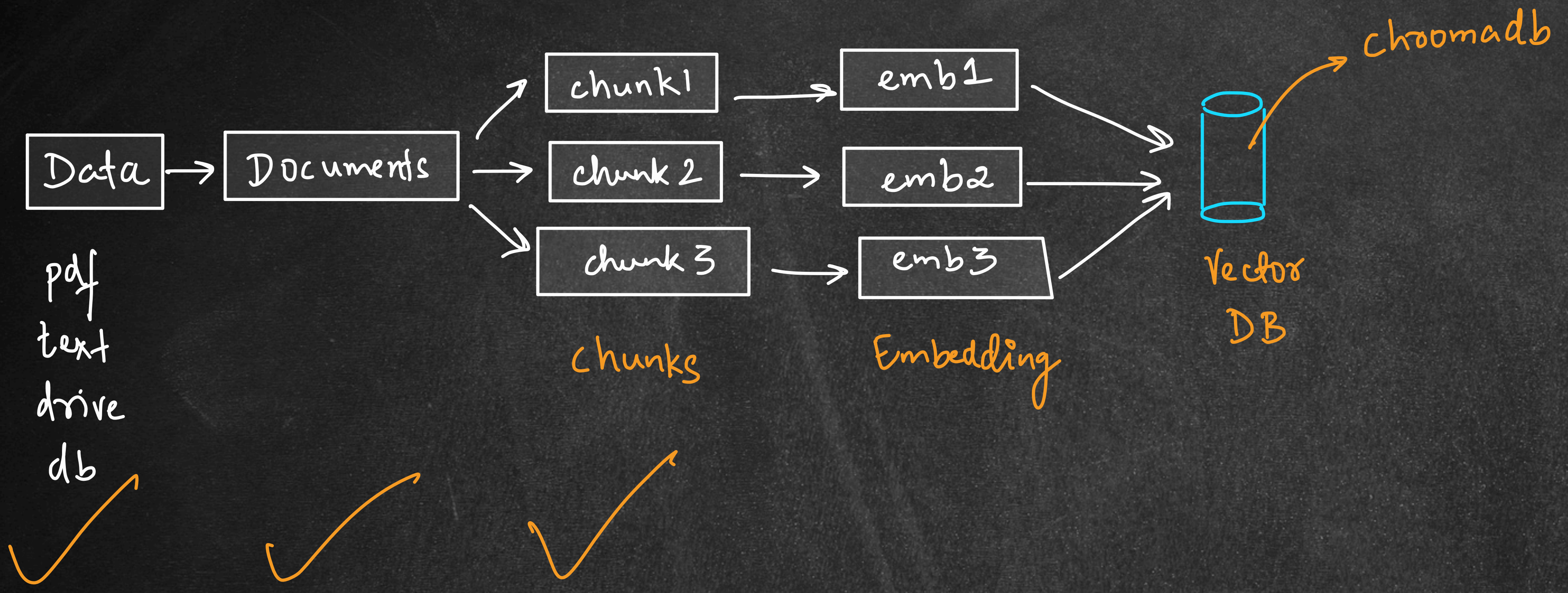
Ingestion Pipeline Implementaion

Chunks



Ingestion Pipeline Implementaion

Embedding



Embedding Manager

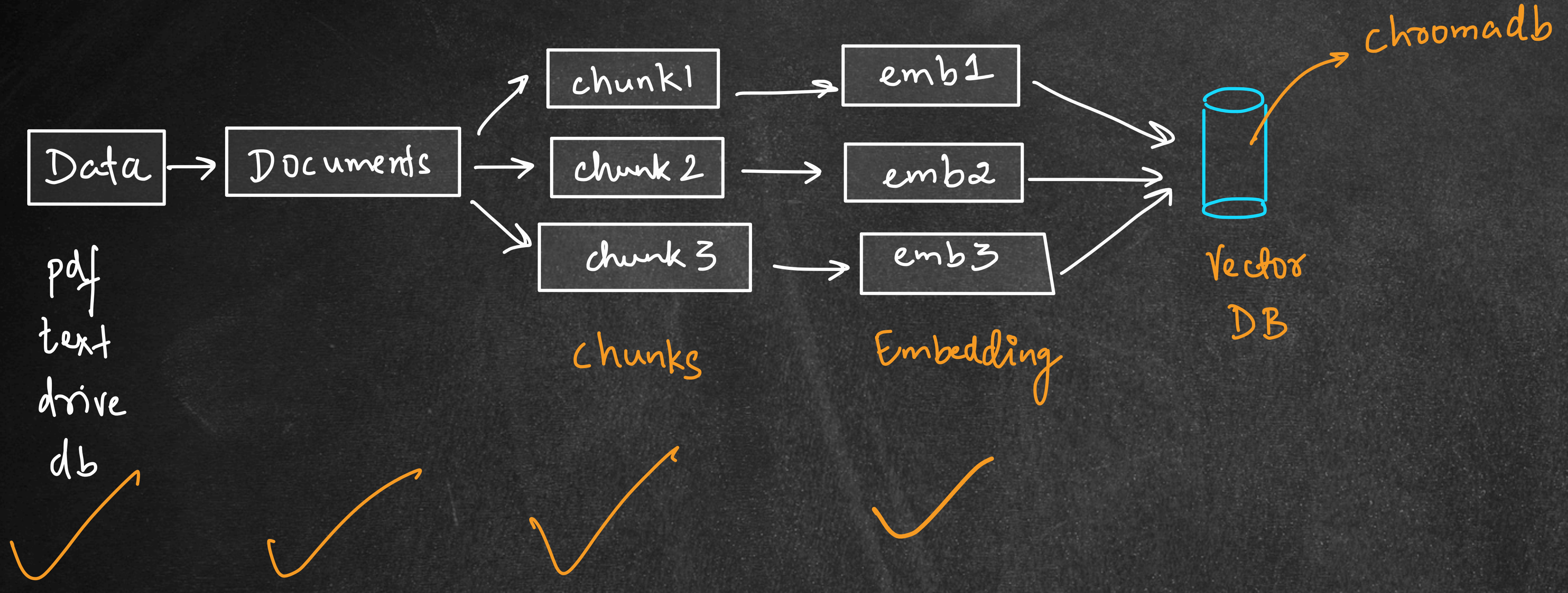
obj

code reusability ↑

Query Embedding

Ingestion Pipeline Implementaion → Vector Store Manager

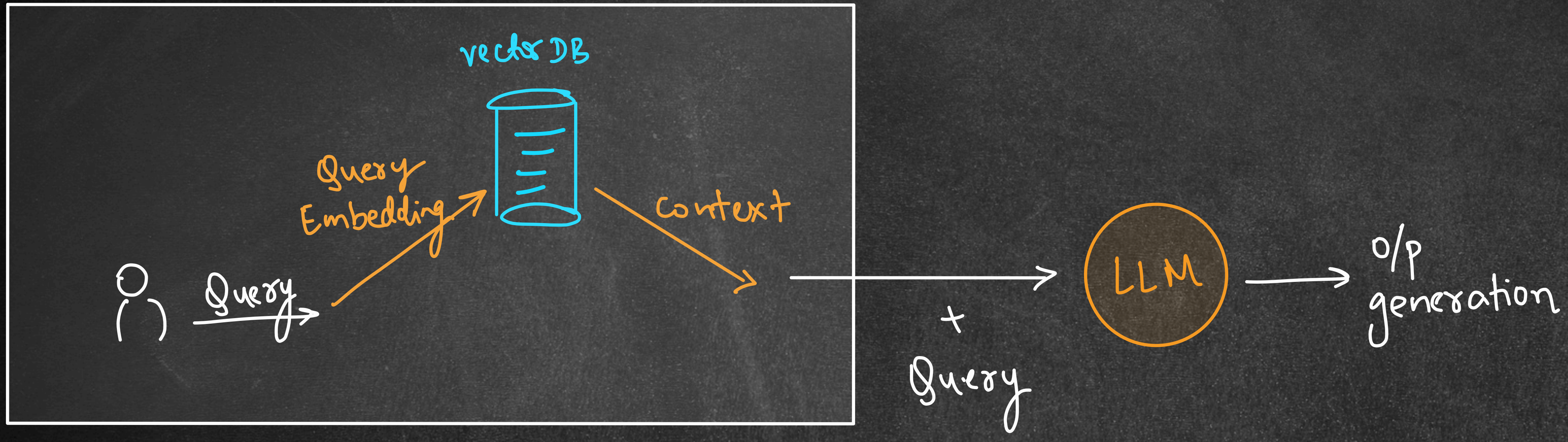
Vector Store



1. initialize vector store - constructor
2. VS_initialize → path, collection
3. store docs → VS

Retrieval Pipeline Implementaion

Retriever



RAG Retriever

return a list of results (context)